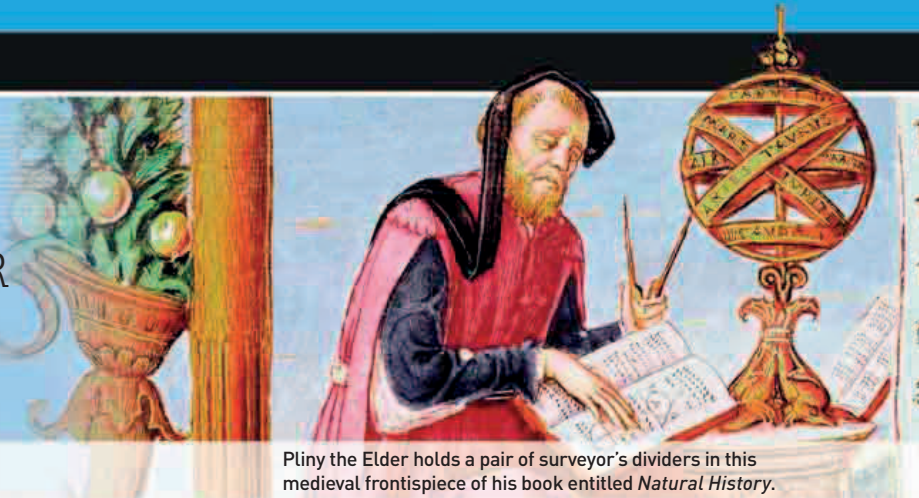


“ BEARS WHEN FIRST BORN ARE SHAPELESS MASSES OF WHITE FLESH A LITTLE LARGER THAN MICE, THEIR CLAWS ALONE BEING PROMINENT. ”

Pliny the Elder, Roman historian and philosopher from *Natural History*, Book VIII, 77



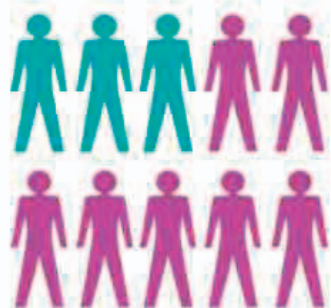
Pliny the Elder holds a pair of surveyor's dividers in this medieval frontispiece of his book entitled *Natural History*.

THE ROMAN HISTORIAN AND PHILOSOPHER PLINY THE ELDER

(23–79) compiled *Natural History*, a 37-volume summary of ancient knowledge, which he completed in 77. It contains much of what we know about Greek and Roman science, covering mineralogy, astronomy, mathematics, geography, and ethnography, as well as including detailed sections on botany and zoology. *Natural History* is also significant because it contains the only references we have to the work of earlier scientists.

During this period three Greek physicians published notable works on **anatomy and diseases**. In the late 1st century, Aretaeus of Cappadocia wrote *The Causes and Signs of Acute and Chronic Diseases*, describing a vast range of diseases, their diagnosis, causes, and treatment. He was the **first physician to describe both diabetes and celiac disease**. Among the other conditions he dealt with were pleurisy, pneumonia, asthma, cholera, and phthisis (tuberculosis), for which he prescribed trips to the seaside.

In 100, Greek physician Rufus of Ephesus wrote *On the Names of the Parts of the Human Body*, summarizing the **Roman knowledge of anatomy**. He gave a detailed description of the eye,



3:10 **Infant mortality rate in Rome**
Despite medical advances, in the 1st and 2nd centuries, the infant mortality rate in Rome was still roughly 30 percent.

and was the **first to identify the optic chiasma**, where the optic nerves partially cross in the brain. He was also the first to name the pancreas and made a detailed study of melancholia (depression).

In the early 2nd century, the Greek physician Soranus of Ephesus produced *On the Diseases of Women*. This was the most comprehensive work on the subject from the ancient world. In it, he described the appropriate training for midwives and gave instructions for managing childbirth, such as the use of the obstetric chair or birthing

stool and how to administer intrauterine injections, and explained the use of the speculum mirror for internal examinations, as well as giving a detailed description of specific gynecological conditions. Soranus of Ephesus also pioneered the **science of pediatrics**. His work contained advice on the early care of infants, including the making of artificial teats for feeding, and accounts of childhood afflictions such as tonsillitis, a variety of fevers, and heatstroke.

Zhang Heng (78–139) was a polymath whose work

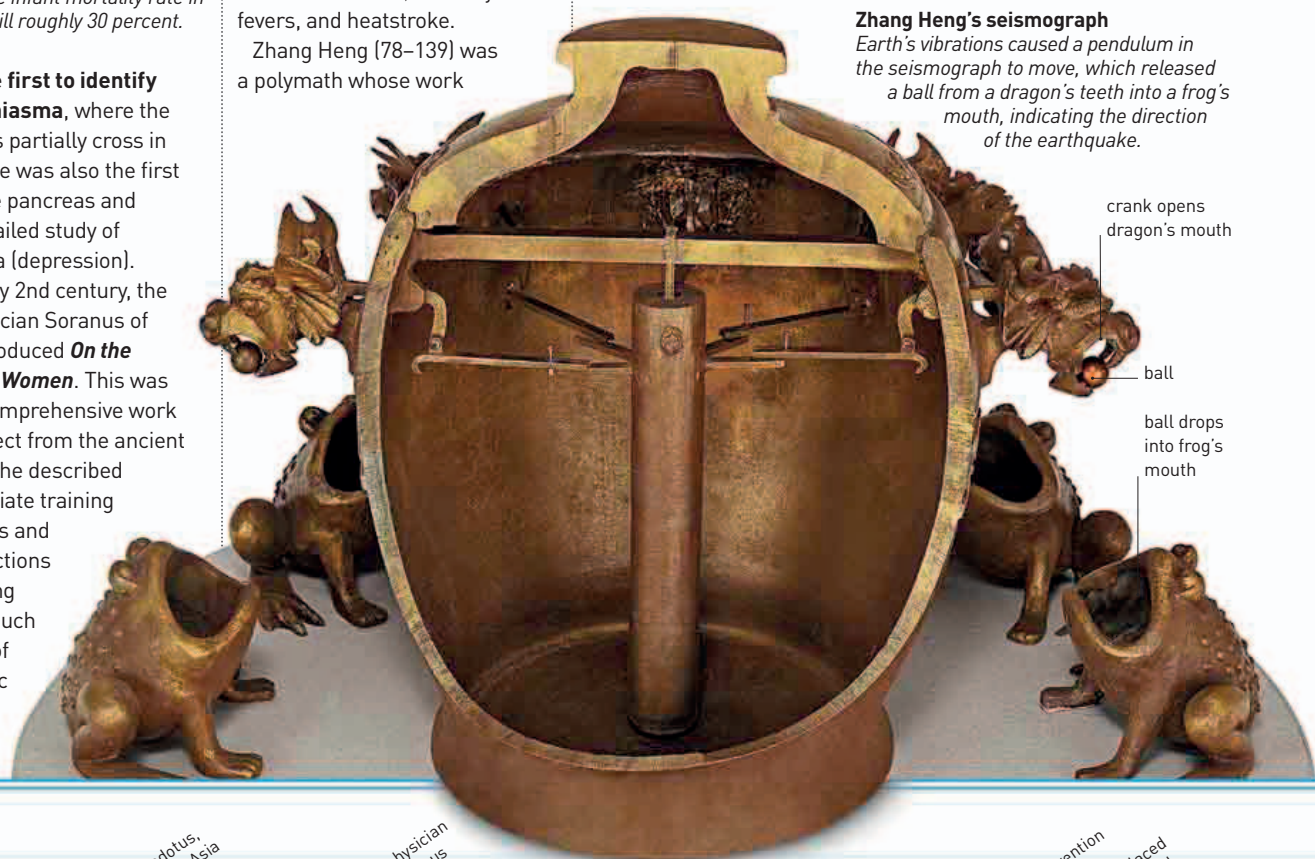
in China in the early 2nd century included calculating a value for pi, the identification of 124 constellations in the sky, and the construction of an armillary sphere with moving parts to show the rotation of the planets. He is best known for the **construction of the earliest seismograph**, which he completed in 132. It consisted of a bronze urn with a pendulum inside. When a tremor occurred, the pendulum swung in the

direction of one of eight dragons' heads, which opened and released a ball into the mouth of a bronze frog below, indicating the direction of the earthquake. In 138, Zhang Heng used the seismograph to **successfully detect an earthquake** that had happened more than 400 miles (640 km) from the Chinese court, where he was demonstrating it.

In the 3rd century BCE, the Romans discovered the principle

Zhang Heng's seismograph

Earth's vibrations caused a pendulum in the seismograph to move, which released a ball from a dragon's teeth into a frog's mouth, indicating the direction of the earthquake.



c.75 Demosthenes Philalithes writes a textbook on ophthalmology

77 Roman historian Pliny the Elder completes his *Natural History*

100 Rufus of Ephesus writes a treatise on anatomy

c.100 Herodotus, a physician of Asia Minor, identifies smallpox

c.100 Greek physician Soranus of Ephesus produces a treatise on gynecology

c.100 Aretaeus of Cappadocia describes diabetes

c.100 Menelaus of Alexandria produces work on trigonometry, including proof of Menelaus's theorem of spherical triangles

After 100 The Treadle loom is developed in China

After 100 Archigenes sets down 10 criteria for measuring the pulse